

Learned model — METHODS

Formula-level spec of every estimator

The formal math for each estimator in `mirna_hallmark/learned/`. This document is the **spec** — notation, objective, formula, and one line of “what it computes” — kept clean of prose. Companions: `LEARNED_MODEL_RATIONALE.md` (the *why* behind every section here — section numbers mirror this doc), `LEARNED_MODEL_ESTIMATOR_MAP.md` (which estimator for which job), and `LEARNED_MODEL_VALIDATION.md` (does it work). The precursor **pressure** model’s math is separate, in `METHODS.md`.

§0 — Notation & identification

Per gene g , aligned on tumour participants (PAM50 Normal-like dropped):

- Y — target mRNA, $\log_2(\text{TPM} + 1)$, vector over participants.
- X — regulator-arm abundance, $\log_2(\text{RPM} + 1)$, participant $\times$ arm (undetected arm $\rightarrow 0$).
- C — confounder block (participant $\times$ conf), see §1.
- w — adaptive-penalty prior weight per arm, see §2.

Candidate regulators (inclusion, migrated 2026-07-06): POOLED-HE = miRTarBase-HE $\cup$ TarBase-v9 low-throughput functional (reporter / western / proteomics, non-weak) — `ledger.pooled_he_edges()`, $\approx 5,940$ edges over 1,571 Hallmark genes; `assemble_gene(he_only=True)` returns this. Replaced miRTarBase-HE-only; gate-stable on anchors ($\Delta\rho \leq 0.003$).

Residualisation (matrix-view identification, Design §2/§B): for a vector v and matrix C ,

$$\text{resid}(v \mid C) = v - C(C^+v)$$

(OLS fit removed; intercept included). Every coupling / weight below is estimated in the residual space $(I - P_C)$ — i.e. C is partialled out of **both** the predictor and the target.

§1 — Confounder block C

Modules: `data.assemble_gene`.

Core (always): $C = [\text{CPE}, \text{target_cn}, \text{mal_prolif}]$.

- CPE — consensus tumour purity (per participant).
- `target_cn` — gene g ’s own ASCAT3 copy number (`_target_cn`), the gene’s genomic dosage.
- `mal_prolif` (`_malignant_prolif`) — malignant-compartment proliferation:

$$\text{score} = \text{mean}_{g' \in (\text{E2FUG2M})} z_{\text{row}}(Y_{g'}), \quad \text{mal_prolif} = \text{resid}(\text{score} \mid \text{CancerEpithelial_fraction})$$

(removes the composition-driven part, leaving per-malignant-cell proliferation).

Optional add-ons (joined as extra C columns):

- `deconv` — the 8 non-malignant CIBERSORTx fractions [CAFs, T, Myeloid, B, Endothelial, PVL, NormalEp (Cancer-Epithelial excluded — conditioning on the target’s own compartment over-controls).
- `mycaf (_mycaf) = resid ( mean  $z$ (ACTA2,TAGLN,POSTN,FAP,COL11A1,THBS2,MYH11,CNN1) | CAF_fraction)` — myCAF-vs-iCAF axis.
- `mal_emt (_mal_emt) = resid ( mean  $z$ (VIM,FN1,CDH2,SNAI2,ZEB1,ZEB2,TWIST1,SPARC) | non-malignant_fractions)` — within-cancer EMT state.
- HRD off; naive batch off (post-fit conditioning only, never in-fit).

Defaults (which C is actually used). The core [CPE, target_cn, mal_prolif] is **always on**. The non-malignant CIBERSORTx fractions (`deconv`) are **opt-in**: OFF for the gate / coupling (`assemble_gene deconv=False`), but **forced ON for the canonical attribution** (`canonical_M / budget_shift / decompose call deconv=True`) — those weights are *cell-intrinsic*. The metagene- z composition (`_state_metagene_cov`, a per-program z -metagene) is only the cross-state fall-back for GTEx (no CIBERSORTx there); NAT / tumour use the CIBERSORTx fractions. So: tumour attribution = fractions; gate / coupling = core only unless `deconv` requested; healthy = metagene- z until the pending GTEx CIBERSORTx run lands.

Post-fit conditioning. `batch` (TCGA plate dummies) is **never in the lasso** — an in-fit ComBat-style correction removes biological heterogeneity along with the batch — so `batch` is appended to C **only** for the partial-Spearman ρ significance computation (`oof_gate`), conditioning the *test*, not the *model*.

Within-cancer vs non-malignant. `mal_prolif` residualises the proliferation metagene on the Cancer-Epithelial fraction $\rightarrow$ per-malignant-cell proliferation; the `deconv` block adds the non-malignant fractions (CAFs / immune / ...) to strip stromal / immune composition. Cancer-Epithelial is deliberately excluded from `deconv` — conditioning on the compartment the target is expressed in over-controls the signal.

§2 — The prior w

Modules: `evidence/ledger.py`, `evidence/methods.py`.

Evidence enters as **magnitude**, non-circularly. Per (edge, assay class) count distinct PMIDs; the fused edge weight is

$$w_m = \sum_{\text{class}} w_{\text{class}} \cdot \log(1 + \# \text{ distinct_PMID}(m, \text{class})).$$

- w_{class} (assay directness) is either **hand-set** (`CLASS_WEIGHT`: reporter 3 > western/proteomics 2.5 > qpcr 1.5 > degradome/chimeric 1 > ago-clip 0.5 > other 0) or **learned** (`CLASS_WEIGHT_MRNA`): non-negative least squares of measured transfection repression (McGeary GSE140217) on per-class log-PMID, within-family-centred.
- $w = \text{clip}(w_m, \geq 10^{-3})$. Alternatives: `scanmir` (K_D affinity), fused (existence $\times$ magnitude).

Default. The `assemble_gene / oof_gate` signature default is `evidence_score` (hand-weighted), but the canonical / reported fits force `ledger` (the weighted, PMID-deduped, method-centric fusion) — so we use **weighted evidence, not binary inclusion**. Empirically the source is non-critical (hub panel, family gate): mean ρ_{model} is `ledger` $-0.345 \approx$ `evidence` $-0.340 \approx$

fused $-0.342 \approx$ scanmir -0.331 ; stability 0.86/0.86/0.85/0.84; ledger passes 5/5 gate, scanmir/fused 4/5 (scanmir breaks GATA3, fused breaks PTEN vs-curated). $\Rightarrow$ ledger is the marginal best and is canonical; the prior sets *ordering*, the data sets *magnitude* (§3).

§3 — The model: gene-focused non-negative adaptive lasso

Modules: *regression.fit_gene*.

Objective (one gene), on C -residualised $\tilde{y} = \text{resid}(Y \mid C)$, $\tilde{X} = \text{resid}(X \mid C)$:

$$\hat{M} = \arg \min_{M \geq 0} \|\tilde{y} - \tilde{X}M\|^2 + \alpha \sum_m \frac{|M_m|}{w_m}.$$

- Non-negativity encodes “miRNAs repress” (Design §Decision C).
- **Adaptive** penalty $\lambda_m = \alpha/w_m$: strong-evidence edges resist shrinkage, weak edges shrink out. The prior is an **inverse penalty** (resistance to being zeroed), **not** a location — it sets ordering / selection, not magnitude.
- Implementation (Zou reparametrisation): scale column m by $w_m \rightarrow$ plain non-negative lasso (uniform α) $\rightarrow$ unscale $M_m = b_m w_m$. Default $\alpha = 0.005$ (fixed; non-critical — global grid 0.001–0.02 gives flat mean $\rho_{\text{model}} \approx -0.340$, gate 8/8; α is a **sparsity** knob [nonzero 9 $\rightarrow$ 7], not a coupling knob; only 0.05 over-shrinks). α touches **only** the lasso — the canonical attribution (§5) is unpenalized bagged NNLS.
- Aggregate repression pressure: $\rho_{\text{agg}} = XM$ (higher $\Rightarrow$ more repression $\Rightarrow$ lower Y).
- This is the model used for coupling / selection / validation (§6) and discovery. Contribution $\approx a \cdot M$ (unsaturated linear occupancy; the saturating $a/(a + K)$ link was tried and failed the gate).

§4 — Seed-family collapse

Modules: *families.collapse_by_family*, Design §F.

Same-seed arms are near-collinear $\Rightarrow$ the identified estimand is family $\rightarrow$ gene (arm = nomination).

$$X_{\text{fam}} = \log_2 \left(1 + \sum_{m \in \text{family}} (2^{x_m} - 1) \right) \quad (\text{true RPM pool, re-logged}), \quad w_{\text{fam}} = \max_{m \in \text{family}} w_m.$$

§5 — Canonical attribution: bagged family NNLS

Modules: *states._bagged_nnls*, *canonical_M*.

The stable readout of per-edge weight (single-lasso coefficients are unstable under collinearity, corr 0.03). On C -residualised, z -scored family predictors:

$$\begin{aligned} y_r &= -\text{resid}(Y \mid C) && (\text{sign: repression} \rightarrow +) \\ X_z &= \text{zscore_cols}(X_{\text{fam}}) && = (X_{\text{fam}} - \text{mean}) / \text{sd} \\ X_z[:, \text{sd} < 0.1] &= 0 && (\text{variance floor}) \\ \text{for } b = 1..B : \text{idx}_b &= \text{bootstrap}(n), \quad c_b = \text{NNLS}(X_z[\text{idx}_b], y_r[\text{idx}_b]) \\ M_{\text{fam}} &= \frac{1}{B} \sum_b c_b && (\text{bagged, } B = 40) \end{aligned}$$

with the sign chosen so regression maps to a positive weight, a **variance floor** that zeroes any arm undetectable in the state ($sd < 0.1$), and $B = 40$ bootstrap replicates whose bagging drives coefficient variance down by $\sim 1/B$.

- Reproducible ($corr \sim 0.99$ across seeds) and recovers curated drivers (PTEN miR-21/103a/181/182/96). z -scored because level-scale NNLS is ill-conditioned (recovers noise arms).
- **arm_level broadcast:** $M_{\text{arm}} = M_{\text{fam}}(\text{family}(\text{arm}))$ — family weight shared to member arms (per-arm split = nomination). Used everywhere coefficients are read: budget, ΔM /wiring, decompose, card.
- **ΔM / cross-state wiring:** fit M_{fam} separately per state (tumour “01” / NAT “11” / GTEx) on the **same fixed** family support $\rightarrow \Delta M = M_{\text{state1}} - M_{\text{state2}}$ (fixed support $\Rightarrow$ comparable estimand).

§5a — Budget contribution (family-size-correct) (`states.budget_shift`)

Realized within-gene budget, apportioned so a multi-arm family isn’t over-counted:

$$\text{contribution}(\text{arm}, \text{state}) = M_{\text{fam}} \cdot X_{\text{fam}}(\text{state}) \cdot \frac{2^{a_{\text{arm}}(\text{state})}}{\sum_{m \in \text{fam}} 2^{a_m(\text{state})}},$$

$$\text{share}(\text{arm}) = \frac{\text{contribution}}{\sum_{\text{arms}} \text{contribution}}, \quad \text{rank} = \text{descending rank of contribution},$$

so $\sum_{\text{arm} \in \text{fam}} \text{contribution} = M_{\text{fam}} \cdot X_{\text{fam}}$ (no family-size inflation). A singleton’s $X_{\text{fam}} =$ its own abundance, so $\text{contribution} = Ma$ (unchanged).

§6 — Validation gate: out-of-fold aggregate coupling

Modules: `mvp.oof_gate`, `mvp.gate_fdr`.

5-fold over participants; fit M on train, score held-out:

$$\text{oof_model}[te] = X[te] M_{\text{train}}, \quad \text{oof_abund}[te] = \text{mean}_m X[te] \quad (\text{per participant})$$

$$\rho_{\text{model}} = \text{spearman}(\text{resid}(\text{oof_model} \mid C), \text{resid}(Y \mid C)) \quad (\text{expect} < 0)$$

$$\rho_{\text{abund}} = \text{spearman}(\text{resid}(\text{oof_abund} \mid C), \text{resid}(Y \mid C)) \quad \text{Bar-1: equal-weight inclusion}$$

$$\rho_{\text{curated}} = \text{spearman}(\text{resid}(Xw_{\text{prior}} \mid C), \text{resid}(Y \mid C)) \quad \text{Bar-2: frozen evidence weights (no fit)}$$

Gate passes if $\rho_{\text{model}} < \rho_{\text{abund}}$ (learning beats abundance) **and** $\rho_{\text{model}} \leq \rho_{\text{curated}}$ ($\geq$ frozen prior). Genome-wide FDR (`gate_fdr`): per gene one-sided partial- t p for $\rho_{\text{model}} < 0$ on $df \approx n - 8$, then BH and Benjamini-Yekutieli across genes (BY for shared-regulator dependence). Only meaningful for ≥ 2 -regulator genes (singletons: $XM \equiv$ abundance up to scaling).

§7 — Per-edge coupling: family-aware permutation partial-Spearman

Modules: `coupling.edge_coupling` $\rightarrow$ `coupling_inference.permutation_pvalues`.

Per gene, predictor rows = arms, target = gene mRNA (repeated), covariates = C , families = seed family:

$$\text{obs}_m = \sum_s \text{unit}(\text{resid}(\text{rank}(x_m) \mid \text{rank}(C)))_s \cdot \text{unit}(\text{resid}(\text{rank}(Y) \mid \text{rank}(C)))_s \quad (\text{partial Spearman})$$

- **null:** the same permutation π of the predictor-residual sample order applied to **every** row (preserves family dependence).

$$p_{\text{perm}} = \frac{1 + \#\{\text{null} \leq \text{obs}\}}{n_{\text{perm}} + 1}, \quad p_z = \Phi\left(\frac{\text{obs} - \mu_{\text{null}}}{\sigma_{\text{null}}}\right) \quad (\text{smooth tail from null moments})$$

$$q_{\text{bh}} = \text{BH}(p_z), \quad q_{\text{by}} = \text{BY}(p_z), \quad q_{\text{family}} = \text{Simes-per-family BH}(p_z) \quad (\S 14)$$

Called **per gene** (C carries per-gene target_cn). Genome-wide: second-level BY across all edges.

§8 — Causality: CN-locus instrument

Modules: *instrument.edge_instrument*, *run_clean*.

miRNA-locus copy number $\text{CN}(m)$ instruments arm dose:

$$\text{first stage: } x_m = \gamma \text{CN}(m) + C\delta + u, \quad F = \frac{(\text{RSS}_{\text{reduced}} - \text{RSS}_{\text{full}})/1}{\text{RSS}_{\text{full}}/\text{df}} \quad \text{usable if } F > 10$$

reduced form: $\rho_{\text{reduced}} = \text{spearman}(\text{resid}(\text{CN} \mid C), \text{resid}(Y \mid C))$ expect < 0 (genetic dose $\downarrow$ target)

observational: $\rho_{\text{obs}} = \text{spearman}(\text{resid}(x_m \mid C), \text{resid}(Y \mid C))$

$$\text{causal_concordant} = (\rho_{\text{reduced}} < 0) \wedge (\rho_{\text{obs}} < 0) \wedge (F > 10)$$

$$p_{\text{reduced}} = \text{one-sided Spearman-}p(\rho_{\text{reduced}} < 0), \quad q_{\text{reduced,by}} = \text{BY}(p_{\text{reduced}})$$

Cluster-exclusion (*cluster_attribution*): CN moves the whole genomic cluster. For co-located miRNAs that also target g (set $S = \{m\} \cup \text{co-targeters}$), condition each on the others $+C$:

$$\text{part}(k) = \text{spearman}(\text{resid}(x_k \mid \{x_{j \neq k}\} \cup C), \text{resid}(Y \mid \{x_{j \neq k}\} \cup C)).$$

$\text{arm_unique} = (\text{part}(m) < -0.1) \wedge (\text{no co-targeter survives})$; $\text{strong_causal} = \text{concordant} \wedge (\text{cluster_clean} \vee \text{arm_unique})$; strong_causal_fdr adds $q_{\text{reduced,by}} < 0.05$.

§9 — Driver resolution inside a family

Modules: *families.resolution_report*.

For a model-selected multi-member family, each member's anti-corr is measured *net of its family-mates* ($+C$):

$$\text{part}(\text{arm}) = \text{spearman}(\text{resid}(x_{\text{arm}} \mid \{\text{mates}\} \cup C), \text{resid}(Y \mid \{\text{mates}\} \cup C)).$$

A member **survives** if $\text{part} < -0.1$ (real conditioned repression). The nomination reflects how the surviving anti-corr distributes (not a forced single winner):

- **0 survivors** → “**family-only**” — the split is unidentified (collinear members lose their residual variance $\Rightarrow \text{part} \rightarrow 0$, auto-fail; e.g. ESR1 miR-221/222).
- **1 survivor** → “**single-driver**” (e.g. PTEN miR-181b, miR-429).
- ≥ 2 **survivors** → “**co-drivers**” — the repression is carried by several members, reported as a set (e.g. PTEN miR-17\$~\$92 = miR-106b -0.22 + miR-20a -0.13 ; ZEB1 = miR-429 + miR-200c).

Columns: resolution (the class), n_drivers, drivers (the surviving set, strongest first), driver (strongest survivor, compat), member_partials (every member’s part). Empirically $\approx 13:7:11$ single : co : family-only.

§10 — Uncertainty: hierarchical Bayes

Modules: *hierarchical_gibbs*.

Per program (genes sharing regulators), on C -residualised z -scored X :

$$\begin{aligned} y_g &= X_g \beta_g + \varepsilon_g, & \varepsilon_g &\sim \mathcal{N}(0, \sigma_g^2) \\ \beta(m, g) &\sim \mathcal{N}(\mu_m, \tau^2) & & \text{(miRNA-level pooling across the genes } m \text{ targets)} \\ \mu_m &\sim \mathcal{N}(0, \omega^2), & \tau^2, \sigma_g^2 &\sim \text{InvGamma} \end{aligned}$$

Gibbs (conjugate): $\beta_g \mid \cdot \sim \mathcal{N}(V(X^\top X/\sigma^2 + \mu/\tau^2), V)$, $V = (X^\top X/\sigma^2 + I/\tau^2)^{-1}$; $\mu_m \mid \cdot$ pooled across its genes; $\tau^2, \sigma_g^2 \mid \cdot$ inverse-gamma. Output: posterior mean $\pm$ sd of β ; **identified edge** = $|\beta/\text{sd}| > 2$. (Cross-gene magnitude pooling does not improve OOF coupling here — used as the uncertainty object; coupling stays per-gene, §6.)

§11 — Subtype tests

Modules: *subtype.py*.

- **Level** (Kruskal): `kruskal_across_strata(XM, PAM50)` — does pressure magnitude differ across subtypes.
- **Slope** (interaction): nested rank-OLS F -test $\text{rank}(Y) \sim \text{rank}(\rho_{\text{agg}}) \cdot \text{subtype} + C$ vs common-slope — does the coupling differ; BH across genes.
- **Wiring, independent** (`subtype_wiring`): fit canonical M_{fam} (§5) separately per subtype; mean pairwise cross-subtype $\text{spearman}(M)$ vs an n -matched noise floor (random cohort subsamples). High variance (differences of two small- n fits) — *over-calls* remodeling.
- **Wiring, pooled** (Decision H, `subtype_wiring_pooled`): whole-cohort M_{all} (§5) is the **prior mean**; per subtype $M_s = M_{\text{all}} + \delta_s$ via ridge-toward- M_{all} .

$$\delta_s = \arg \min_{\delta} \left\| (y_r - X_z M_{\text{all}})[s] - X_z[s] \delta \right\|^2 + \lambda \|\delta\|^2 \quad (\text{closed form; } \lambda = 10).$$

Δ is regularised $\rightarrow$ small- n noise shrinks to 0; remodeling = $\|\delta_s\|_1$ vs an n -matched null. **Corrects the independent version**: PTEN borderline ($\sim 1.35\times$ across $\lambda = 1\text{--}30$, not robust),

only RB1-Her2 modest ($\sim 1.5\times$) $\Rightarrow$ subtype wiring mostly conserved. This is the design's $M_T = M_H + \Delta$ nesting, subtype flavour (state flavour: same with M_H from GTEx as the prior mean).

§12 — Discovery

Modules: *discovery.py*.

Orphan edge (m, g) beyond the curated POOLED-HE regulators:

$$\text{spearman}(\text{resid}(x_m \mid C \cup \text{pooledHE-aggregate}), \text{resid}(Y \mid \cdot)) < \text{min_partial},$$

gated by scanMiR biochemical support and (deconv) composition retention. Edge-level FDR from a permutation null-scan (real vs permuted hits). Family-level: collapse to seed-family $\rightarrow$ gene hypotheses (§14).

§13 — Enrichment

Modules: *enrichment.py*, *stats.hypergeom_enrichment*.

For a target set (discoveries / a hub miRNA's targets) vs Hallmark program P over the HE universe U :

$$\text{fold} = \frac{k/|P|}{K/|U|}, \quad p = P(X \geq k) \text{ hypergeometric with } (|U|, K, |P|); \quad \text{BH across programs.}$$

§14 — Multiple-testing machinery

Modules: *stats.bh_fdr*, *coupling_inference.{benjamini_yekutieli, simes_p, family_simes_fdr}*.

- **BH:** $q_{(i)} = \min_{j \geq i} p_{(j)} n/j$ (step-up).
- **Benjamini-Yekutieli** (arbitrary dependence): BH with α/H_n , $H_n = \sum_{i=1}^n 1/i$.
- **Simes per family:** family $p = \min_i p_{(i)} k/i$ over the family's k tests; BH across families $\rightarrow$ per-test q .
- **Family-wise min-statistic null:** per draw take the family's most-extreme null statistic; $p_{\text{family}} = (1 + \#\{\text{null}_{\min} \leq \text{obs}_{\min}\})/(n_{\text{perm}} + 1)$.

Hypothesis unit = seed family (Design §F): family-mate arms are **one** test, not N .

§15 — Identity vs magnitude attribution

Modules: *attribution.identity_vs_magnitude*, *Design §Decision I*.

Over the gene's nonzero seed-families, two objects — never conflated:

- **Magnitude** (budget) = $M_{\text{fam}} X_{\text{fam}}$ normalised (§5a) — realized force (abundance-included).
- **Identity** = **Shapley** credit for the fixed-weight aggregate's explained variance:

$$\varphi_f = \text{mean}_{\text{orderings } \pi} [R^2(S \cup \{f\}) - R^2(S)], \quad S = f\text{'s predecessors in } \pi,$$

$$\text{pred}_S = \sum_{f' \in S} M_{f'} \tilde{X}_{f'}, \quad R^2(S) = \text{corr}(\text{pred}_S, \tilde{y})^2,$$

with $\tilde{X}/\tilde{y}$ C -residualised ($\tilde{X}$ z -scored). Sampled orderings ($n_{\text{perm}} = 400$; = **LMG** for the linear aggregate — exact = 2^p). identity_share = $\varphi_f / \sum \varphi$.

- gap = identity — magnitude: + = explains more of Y than its budget (quiet on-target owner, e.g. ESR1 miR-18: identity 0.54 / budget 0.20); — = high budget but doesn't explain Y (loud passenger, e.g. ESR1 miR-22: budget 0.26 / identity 0.01). Budget rewards abundance ($M\tilde{X}$); identity rewards on-target coupling — the gap is *primarily* abundance-vs-coupling, and *secondarily* collinearity (co-varying families split the shared R^2).
- **Discipline:** identity $\perp$ coupling (§6/§7) — the share distributes the coupling that *exists*; it is never evidence a driver exists. This retires the precursor softmax-share identity.

§16 — Structural identity: the loss lens

Modules: *structural_identity.py*, Design §Decision I loss variant.

Expression-free “who is **designed** to specifically repress g ”, so a *silenced* specialist (M / budget / Shapley all 0 — no variance, no abundance) still surfaces. Candidate families = pooled-HE (§0) $\cap$ scanMiR-modelled arms, with a TargetScan-context++ fallback for arms scanMiR lacks (recovers e.g. miR-137). Per seed-family f of g :

- $\text{potential}(f, g) = \text{biochem strength}(\text{scanMiR} \mid \text{repression} \mid \text{TargetScan ts_mag}) \times (1 + \log(1 + \text{evidence ledger}))$, direction-filtered (drop TarBase-v9 validated-activating). Family = its strongest member.
- $\text{specificity}(f, g) = \frac{\sum_{\text{members}} \text{ev}(\cdot, g)}{\sum_{\text{members}} \sum_{g'} \text{ev}(\cdot, g')}$ — the family's pooled evidence-mass fraction aimed at g (pool the whole seed family, not one representative arm — else a thinly-studied member fakes concentration). There is no biochemical specialist (every arm hits ~ 700 targets $\sim$ uniformly), so specificity is purely an evidence construct — no biochemical floor.
- $\text{confidence}(f) = \min(1, \text{pooled_ev_depth}(f)/10)$, $\text{ev_depth} = \#$ genes with any family evidence.
- $\text{structural}(f, g) = \text{potential} \cdot \text{specificity} \cdot \text{confidence}$ (confidence a **multiplier**: no evidence $\Rightarrow$ no ownership). structural_share = gene-local normalisation.

lost_specialist(gene) overlays baseline-active (GTEx/NAT $> \log_2(11)$) $\wedge$ tumour-silenced ($< \text{floor}$) + §17.

§17 — Loss/gain mechanism: promoter methylation gate

Modules: *methylation.py*.

Tumour — normal promoter $\Delta\beta$ over CpG probes directly overlapping the miRNA promoter (full sesame HM450 manifest, 485,577 probes; window [TSS — 2000, TSS + 500] around the pri-miRNA hairpin; **no cCRE hop** — the gene-centric probe annotation misses miRNA loci, which is why the pipeline fell back to cCRE). Group means over 802 tumour vs 99 normal raw betas (no barcode map needed). **Bidirectional**:

- $\text{hyper} = (\Delta\beta \geq +0.15) \wedge (\beta_{\text{tumour}} \geq 0.20) \rightarrow \text{arm } \mathbf{\text{silenced}} \rightarrow \text{repression } \mathbf{\text{lost}}$ (target $\uparrow$).
- $\text{hypo} = (\Delta\beta \leq -0.15) \wedge (\beta_{\text{normal}} \geq 0.20) \rightarrow \text{arm } \mathbf{\text{de-repressed}} \rightarrow \text{repression } \mathbf{\text{gained}}$ (target $\downarrow$).

Validated: TSG-miRNAs mean $\Delta\beta + 0.061$ vs oncomiRs -0.070 (miR-124 $+0.193$, 48 CGI probes; miR-21 -0.176). Promoter-probe-union betas cached at `output/learned/mirna_promoter_betas.par`.

Every formula above has a runnable implementation at the cited module.function; see the reproduce index in `LEARNED_MODEL_VALIDATION.md`.